

Student Management System

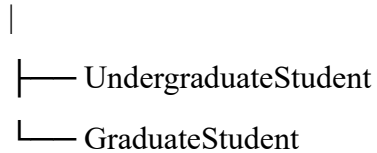
Task

Create a Student Management System using Python OOP concepts.

Required Classes

Create the following classes:

Student



1. Student Class

Create a Student class with the following attributes:

- name
- student_id
- email
- age
- department

Create the following methods:

- display_info()
- calculate_result()
- get_student_type()

2. UndergraduateStudent Class

Inherit from Student.

Add at least one new attribute, such as:

- semester

Override at least one method, such as:

- get_student_type()

3. GraduateStudent Class

Inherit from Student.

Add at least one new attribute, such as:

- research_topic

Override at least one method, such as:

- `get_student_type()`

OOP Concepts Required

Your project must demonstrate:

- Class & Object — Create objects from the classes.
- Attributes — Use student-related attributes.
- Methods — Implement methods for student operations.
- Inheritance — UndergraduateStudent and GraduateStudent inherit from Student.
- Polymorphism — Call the same method on different student objects and get different behavior.
- Method Overriding — Override a parent method in child classes.
- Method Overloading — Use default arguments, `*args`, or `**kwargs` to allow a method to work with different numbers of arguments.
- Encapsulation — Use a private attribute such as `__email` or `__marks`.

Submission Instructions:

- Submit the github repository link

NOTE: You can add any extra property/field to the classes if you want. You must design and implement the methods in your own way. This is just a small assignment to help make the OOP concepts (inheritance, polymorphism, method overriding, encapsulation) clear.